

Groups and Lie Algebras: Solutions to Exercise Sheet 5

1 Two-dimensional lattices

Problem 1. Consider the group $\mathbb{Z}^2$ encoding the translation symmetry of a lattice. Its group of automorphisms is $\text{GL}(2, \mathbb{Z})$, the group of integer matrices with integer inverse. The point groups of the rhombic and rectangular lattices are both abstractly isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$.

- (a) Compute the group homomorphism

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \longrightarrow \text{GL}(2, \mathbb{Z})$$

describing the full symmetry group as a semidirect product in both cases.

- (b) Show that the full symmetry groups of the rhombic and rectangular lattices are not isomorphic.

General setup: lattices and semidirect products

Let $\Lambda \subset \mathbb{R}^2$ be a lattice with basis vectors a_1, a_2 . We identify the lattice with $\mathbb{Z}^2$ by writing

$$m = (m_1, m_2) \in \mathbb{Z}^2 \quad \longleftrightarrow \quad m_1 a_1 + m_2 a_2 \in \Lambda.$$

A point group symmetry is an orthogonal transformation of $\mathbb{R}^2$ which maps the lattice to itself. Since it maps lattice basis vectors to integer linear combinations of lattice basis vectors, it defines an automorphism of $\Lambda \cong \mathbb{Z}^2$. Hence a point group action is described by a group homomorphism

$$\varphi : P \longrightarrow \text{GL}(2, \mathbb{Z}),$$

where P is the point group.

The full lattice symmetry group is then the semidirect product

$$\Lambda \rtimes_{\varphi} P.$$

As a set this is $\mathbb{Z}^2 \times P$, but the multiplication depends on the action φ :

$$(m, p)(n, q) = (m + \varphi(p)n, pq).$$

This formula is the algebraic expression of the fact that point symmetries act on translations. In general, a translation followed by a reflection or rotation is not the same as doing them in the opposite order.

We shall use the following convention for matrices. If $M \in \text{GL}(2, \mathbb{Z})$, then the first column of M is the image of the first basis vector e_1 , and the second column is the image of e_2 . Equivalently,

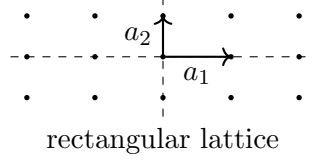
$$Me_j = \text{the } j\text{-th column of } M.$$

Rectangular lattice

Choose rectangular basis vectors a_1, a_2 with

$$a_1 \perp a_2, \quad |a_1| \neq |a_2|.$$

A picture is:



The generic point group contains the identity, reflections in the two coordinate axes, and their product, the rotation by π . Abstractly it is $\mathbb{Z}_2 \times \mathbb{Z}_2$. Let us denote two generators by r_1, r_2 , where

$$r_1(a_1) = -a_1, \quad r_1(a_2) = a_2,$$

and

$$r_2(a_1) = a_1, \quad r_2(a_2) = -a_2.$$

In the basis (a_1, a_2) , the corresponding matrices are

$$\varphi_{\text{rect}}(r_1) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \varphi_{\text{rect}}(r_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These matrices satisfy

$$\varphi_{\text{rect}}(r_1)^2 = \varphi_{\text{rect}}(r_2)^2 = I, \quad \varphi_{\text{rect}}(r_1)\varphi_{\text{rect}}(r_2) = \varphi_{\text{rect}}(r_2)\varphi_{\text{rect}}(r_1),$$

so this indeed defines a homomorphism

$$\varphi_{\text{rect}} : \mathbb{Z}_2 \times \mathbb{Z}_2 \longrightarrow \text{GL}(2, \mathbb{Z}).$$

Its image is

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \right\}.$$

Therefore the full rectangular lattice symmetry group is

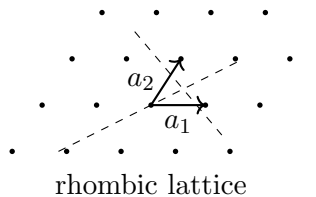
$$G_{\text{rect}} = \mathbb{Z}^2 \rtimes_{\varphi_{\text{rect}}} (\mathbb{Z}_2 \times \mathbb{Z}_2).$$

Rhombic lattice

Now choose a rhombic basis a_1, a_2 with

$$|a_1| = |a_2|, \quad \angle(a_1, a_2) \neq 90^\circ, 60^\circ.$$

A picture is:



The generic point group is again abstractly $\mathbb{Z}_2 \times \mathbb{Z}_2$. However, it acts on $\mathbb{Z}^2$ differently. The two reflections are naturally reflections in the diagonals of the rhombus.

Let s_1 be the reflection which exchanges the two basis vectors:

$$s_1(a_1) = a_2, \quad s_1(a_2) = a_1.$$

In the basis (a_1, a_2) , this is

$$\varphi_{\text{rhomb}}(s_1) = S := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Let s_2 be the reflection in the other diagonal. It acts by

$$s_2(a_1) = -a_2, \quad s_2(a_2) = -a_1,$$

so

$$\varphi_{\text{rhomb}}(s_2) = -S = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

Again these matrices have order two and commute:

$$S^2 = I, \quad (-S)^2 = I, \quad S(-S) = (-S)S = -I.$$

Thus we get a homomorphism

$$\varphi_{\text{rhomb}} : \mathbb{Z}_2 \times \mathbb{Z}_2 \longrightarrow \text{GL}(2, \mathbb{Z})$$

whose image is

$$\{I, -I, S, -S\}.$$

The full rhombic lattice symmetry group is therefore

$$G_{\text{rhomb}} = \mathbb{Z}^2 \rtimes_{\varphi_{\text{rhomb}}} (\mathbb{Z}_2 \times \mathbb{Z}_2).$$

Why the two full symmetry groups are not isomorphic

Although the point groups are abstractly the same, the full symmetry groups are not. The reason is that the point group acts differently on the translation lattice.

Using abelianization

A clean way to prove that the two semidirect products are not isomorphic is to compare their abelianizations. Recall that the abelianization of a group G is

$$G_{\text{ab}} = G/[G, G].$$

Isomorphic groups have isomorphic abelianizations. Therefore, if the abelianizations are different, the groups cannot be isomorphic.

Let

$$G = \mathbb{Z}^2 \rtimes_{\varphi} P,$$

where $P \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ is abelian. In the abelianization, all elements must commute. In particular, a translation $m \in \mathbb{Z}^2$ must commute with an element $p \in P$. But in the semidirect product we have the relation

$$pmp^{-1} = \varphi(p)m.$$

After abelianizing, the left-hand side becomes just m , so we impose

$$\varphi(p)m = m.$$

Equivalently,

$$\varphi(p)m - m = 0.$$

Thus all elements of the form $\varphi(p)m - m$ are killed in the translation subgroup. Hence we define

$$L_\varphi = \langle \varphi(p)m - m \mid p \in P, m \in \mathbb{Z}^2 \rangle.$$

More explicitly, the image of the translation subgroup $\mathbb{Z}^2$ in G_{ab} is obtained from $\mathbb{Z}^2$ by quotienting out precisely these relations. By the first isomorphism theorem, this image is therefore

$$\mathbb{Z}^2 / L_\varphi.$$

This is the translation part of the abelianization.

Since P is already abelian, its image in the abelianization is still P . Moreover, after imposing the relations above, the action of P on the remaining translation part becomes trivial. Therefore the semidirect product becomes an ordinary direct product:

$$G_{\text{ab}} \cong (\mathbb{Z}^2 / L_\varphi) \times P.$$

For the rectangular lattice, using

$$\varphi_{\text{rect}}(r_1) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \varphi_{\text{rect}}(r_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

we find

$$\varphi_{\text{rect}}(r_1)a_1 - a_1 = -2a_1, \quad \varphi_{\text{rect}}(r_2)a_2 - a_2 = -2a_2.$$

Hence

$$L_{\text{rect}} = 2\mathbb{Z}a_1 \times 2\mathbb{Z}a_2,$$

and therefore

$$\mathbb{Z}^2 / L_{\text{rect}} \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

Thus

$$(G_{\text{rect}})_{\text{ab}} \cong (\mathbb{Z}_2 \times \mathbb{Z}_2) \times (\mathbb{Z}_2 \times \mathbb{Z}_2) \cong (\mathbb{Z}_2)^4.$$

For the rhombic lattice, the relevant matrices are

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad -S = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

We compute

$$Sa_1 - a_1 = a_2 - a_1, \quad (-S)a_1 - a_1 = -a_2 - a_1.$$

Thus L_{rhomb} contains

$$a_2 - a_1, \quad a_1 + a_2.$$

These two elements generate L_{rhomb} . Therefore, in the quotient $\mathbb{Z}^2 / L_{\text{rhomb}}$, we have the relations

$$a_2 = a_1, \quad a_1 + a_2 = 0.$$

Combining them gives

$$2a_1 = 0.$$

Hence

$$\mathbb{Z}^2/L_{\text{rhom}} \cong \mathbb{Z}_2.$$

Therefore

$$(G_{\text{rhom}})_{\text{ab}} \cong \mathbb{Z}_2 \times (\mathbb{Z}_2 \times \mathbb{Z}_2) \cong (\mathbb{Z}_2)^3.$$

Since

$$(G_{\text{rect}})_{\text{ab}} \cong (\mathbb{Z}_2)^4, \quad (G_{\text{rhom}})_{\text{ab}} \cong (\mathbb{Z}_2)^3,$$

the abelianizations are not isomorphic. Consequently,

$$G_{\text{rect}} \not\cong G_{\text{rhom}}.$$

This proves that even though the point groups are both abstractly $\mathbb{Z}_2 \times \mathbb{Z}_2$, the full symmetry groups are different because the action on the translation lattice is different.

Using induced action

Another way to prove that the two symmetry groups are not isomorphic is to compare the induced actions on the translation lattice.

Let

$$G_{\text{rect}} = \mathbb{Z}^2 \rtimes_{\varphi_{\text{rect}}} P, \quad G_{\text{rhom}} = \mathbb{Z}^2 \rtimes_{\varphi_{\text{rhom}}} P,$$

where

$$P \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

The translation subgroup $\mathbb{Z}^2$ is characteristic in these lattice symmetry groups. Hence any isomorphism

$$G_{\text{rect}} \cong G_{\text{rhom}}$$

would have to send the translation subgroup of G_{rect} to the translation subgroup of G_{rhom} . Therefore it would induce a change of basis

$$A \in \text{GL}_2(\mathbb{Z})$$

on the translation lattice. In particular, the two actions

$$\varphi_{\text{rect}}, \varphi_{\text{rhom}} : P \rightarrow \text{GL}_2(\mathbb{Z})$$

would have to be conjugate over $\mathbb{Z}$.

For the rectangular lattice, the point group action is generated by

$$\varphi_{\text{rect}}(r_1) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \varphi_{\text{rect}}(r_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The +1-eigenspace of R_1 is generated by e_2 , while the +1-eigenspace of R_2 is generated by e_1 . Thus the primitive fixed directions of these two reflections generate the whole lattice:

$$\langle a_1, a_2 \rangle = \mathbb{Z}^2.$$

Equivalently, they span a sublattice of index 1.

For the rhombic lattice, the point group action is generated by

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad -S = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

The $+1$ -eigenspace of S is generated by

$$a_1 + a_2,$$

while the $+1$ -eigenspace of $-S$ is generated by

$$a_1 - a_2.$$

These two primitive fixed directions generate the sublattice

$$\langle a_1 + a_2, a_1 - a_2 \rangle \subseteq \mathbb{Z}^2.$$

The index of this sublattice is

$$\left| \det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \right| = 2.$$

Thus, in the rectangular case, the two primitive fixed directions generate a sublattice of index 1, whereas in the rhombic case they generate a sublattice of index 2.

This index is invariant under change of basis by an element of $\text{GL}_2(\mathbb{Z})$. Therefore the two point group actions are not conjugate over $\mathbb{Z}$. Hence the semidirect products

$$\mathbb{Z}^2 \rtimes_{\varphi_{\text{rect}}} P \quad \text{and} \quad \mathbb{Z}^2 \rtimes_{\varphi_{\text{rhomb}}} P$$

are not isomorphic.

2 Molecules and selection rules

Problem 2. Consider the water molecule H_2O . Its symmetry group is C_{2v} in Schönflies notation. Find selection rules for the matrix elements of the operators x, y, z . Determine the symmetry group of the benzene molecule C_6H_6 .

The group C_{2v}

We use coordinates as in the lecture: the principal C_2 -axis is the z -axis, and the two reflection planes are the xz - and yz -planes. The group elements are

$$C_{2v} = \{E, C_2, \sigma_{xz}, \sigma_{yz}\}.$$

Their action on coordinates is

$$C_2 : (x, y, z) \mapsto (-x, -y, z),$$

$$\sigma_{xz} : (x, y, z) \mapsto (x, -y, z), \quad \sigma_{yz} : (x, y, z) \mapsto (-x, y, z).$$

The character table is

C_{2v}	E	C_2	σ_{xz}	σ_{yz}
A_1	1	1	1	1
A_2	1	1	-1	-1
B_1	1	-1	1	-1
B_2	1	-1	-1	1

Now determine how the coordinate operators transform. The operator x picks up the same signs as the coordinate function x . Hence

$$x : (1, -1, 1, -1),$$

which is the character of B_1 . Similarly,

$$y : (1, -1, -1, 1),$$

which is B_2 , and

$$z : (1, 1, 1, 1),$$

which is A_1 . Thus

$$x \sim B_1, \quad y \sim B_2, \quad z \sim A_1.$$

General selection rule

Let $|\phi\rangle$ transform in an irreducible representation Γ_i , let $|\psi\rangle$ transform in Γ_f , and let an operator A transform in Γ_A . The matrix element is

$$\langle\psi|A|\phi\rangle.$$

It can be non-zero only if the tensor product

$$\Gamma_f^* \otimes \Gamma_A \otimes \Gamma_i$$

contains the trivial representation. A nonzero matrix element must be invariant under the symmetry group, therefore this tensor product must contain the trivial representation.

For C_{2v} , all irreducible representations are one-dimensional and real, so $\Gamma_f^* = \Gamma_f$. Therefore the condition becomes

$$\Gamma_f \otimes \Gamma_A \otimes \Gamma_i \supset A_1.$$

Because all irreducible representations are one-dimensional, this is equivalent to

$$\Gamma_f \otimes \Gamma_i = \Gamma_A.$$

So the final and initial states must differ by the representation type of the operator.

The multiplication table for the irreducible representations is

$\otimes$	A_1	A_2	B_1	B_2
A_1	A_1	A_2	B_1	B_2
A_2	A_2	A_1	B_2	B_1
B_1	B_1	B_2	A_1	A_2
B_2	B_2	B_1	A_2	A_1

Selection rules for x, y, z

Since

$$x \sim B_1, \quad y \sim B_2, \quad z \sim A_1,$$

we get the following selection rules.

The operator z . Since $z \sim A_1$, the matrix element

$$\langle\psi|z|\phi\rangle$$

can be non-zero only if

$$\Gamma_f \otimes \Gamma_i = A_1.$$

This means

$$\Gamma_f = \Gamma_i.$$

Thus z only connects states of the same symmetry type:

$$A_1 \leftrightarrow A_1, \quad A_2 \leftrightarrow A_2, \quad B_1 \leftrightarrow B_1, \quad B_2 \leftrightarrow B_2.$$

The operator x . Since $x \sim B_1$, we need

$$\Gamma_f \otimes \Gamma_i = B_1.$$

From the multiplication table, the allowed pairs are

$$A_1 \leftrightarrow B_1, \quad A_2 \leftrightarrow B_2.$$

Equivalently,

initial symmetry	allowed final symmetry for x
A_1	B_1
A_2	B_2
B_1	A_1
B_2	A_2

The operator y . Since $y \sim B_2$, we need

$$\Gamma_f \otimes \Gamma_i = B_2.$$

The allowed pairs are

$$A_1 \leftrightarrow B_2, \quad A_2 \leftrightarrow B_1.$$

Equivalently,

initial symmetry	allowed final symmetry for y
A_1	B_2
A_2	B_1
B_1	A_2
B_2	A_1

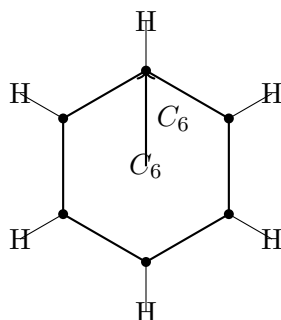
Putting all three cases together:

operator	allowed symmetry pairs
$x \sim B_1$	$A_1 \leftrightarrow B_1, \quad A_2 \leftrightarrow B_2$
$y \sim B_2$	$A_1 \leftrightarrow B_2, \quad A_2 \leftrightarrow B_1$
$z \sim A_1$	$A_1 \leftrightarrow A_1, \quad A_2 \leftrightarrow A_2, \quad B_1 \leftrightarrow B_1, \quad B_2 \leftrightarrow B_2$

These are the selection rules for the dipole operators x, y, z in the water molecule.

The symmetry group of benzene

The benzene molecule C_6H_6 is planar. The six carbon atoms form a regular hexagon, and the six hydrogen atoms lie along the same radial lines. A schematic picture is:



The molecule has a principal sixfold rotation axis perpendicular to the plane of the molecule. This gives rotations by

$$60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ.$$

It also has six twofold axes lying in the plane of the molecule. In addition, because the molecule is planar, reflection in the molecular plane is a symmetry. This is the horizontal reflection σ_h . There are also vertical reflection planes through the principal axis.

Thus the point group is

$$D_{6h}.$$

This is the standard Schönflies point group of benzene.

One way to remember this is:

- D_6 means: a principal C_6 -axis plus six C_2 -axes perpendicular to it;
- the subscript h means that there is a horizontal reflection plane, namely the plane of the molecule.

Therefore benzene has point group

$$\boxed{D_{6h}}.$$

Unlike crystal lattices, molecules do not possess translation symmetry. Therefore the full spatial symmetry group of benzene coincides with its point group:

$$\text{Sym}(\text{benzene}) \cong D_{6h}.$$

3 One-dimensional band structure from translation symmetry

Problem 3. Consider a particle on a one-dimensional lattice with sites labelled by $n \in \mathbb{Z}$. The Hilbert space has an orthonormal basis $\{|n\rangle\}_{n \in \mathbb{Z}}$, and the Hamiltonian is

$$H = -t \sum_{n \in \mathbb{Z}} (|n\rangle\langle n+1| + |n+1\rangle\langle n|), \quad t > 0.$$

- Define the translation operator T by $T|n\rangle = |n+1\rangle$. Show that T is unitary and that $[H, T] = 0$. Conclude that H can be diagonalized in a basis of eigenvectors of T .
- The group generated by T is isomorphic to $\mathbb{Z}$. Its irreducible unitary representations are labelled by $k \in [-\pi, \pi)$ and given by $\chi_k(n) = e^{ikn}$. Define the corresponding states

$$|k\rangle = \sum_{n \in \mathbb{Z}} e^{ikn} |n\rangle.$$

Show that $T|k\rangle = e^{ik}|k\rangle$.

- Compute $H|k\rangle$ and show that $|k\rangle$ is an eigenvector of H with eigenvalue

$$E(k) = -2t \cos k.$$

- Plot $E(k)$ for $k \in [-\pi, \pi]$.

Part (a): the translation operator is unitary

The Hilbert space is $\ell^2(\mathbb{Z})$, with orthonormal basis

$$\langle m | n \rangle = \delta_{mn}.$$

The translation operator is defined by

$$T|n\rangle = |n+1\rangle.$$

Its inverse is

$$T^{-1}|n\rangle = |n-1\rangle.$$

Indeed,

$$TT^{-1}|n\rangle = T|n-1\rangle = |n\rangle, \quad T^{-1}T|n\rangle = T^{-1}|n+1\rangle = |n\rangle.$$

So T^{-1} really is the inverse of T .

To show that T is unitary, we can show that it preserves inner products:

$$\langle Tm | Tn \rangle = \langle m+1 | n+1 \rangle = \delta_{m+1, n+1} = \delta_{mn} = \langle m | n \rangle.$$

Therefore T is unitary. Equivalently,

$$T^\dagger = T^{-1}.$$

Part (a): the Hamiltonian commutes with translations

First compute how H acts on a basis state. Since

$$|n\rangle\langle n+1||m\rangle = \delta_{n+1, m}|n\rangle, \quad |n+1\rangle\langle n||m\rangle = \delta_{n, m}|n+1\rangle,$$

we get

$$H|m\rangle = -t(|m-1\rangle + |m+1\rangle).$$

This is the nearest-neighbour hopping Hamiltonian: from the site m , the particle can hop to $m-1$ or $m+1$.

Now compare $HT|m\rangle$ and $TH|m\rangle$. We have

$$HT|m\rangle = H|m+1\rangle = -t(|m\rangle + |m+2\rangle),$$

while

$$TH|m\rangle = T(-t(|m-1\rangle + |m+1\rangle)) = -t(|m\rangle + |m+2\rangle).$$

Thus

$$HT|m\rangle = TH|m\rangle$$

for every basis vector $|m\rangle$. Therefore

$$[H, T] = 0.$$

This is the algebraic expression of translation symmetry. The Hamiltonian looks the same at every lattice site, so shifting the entire system by one site does not change the physics.

Since H commutes with T , the Hamiltonian preserves the eigenspaces of T . Therefore we can diagonalize H in sectors labelled by irreducible representations of the translation group $\mathbb{Z}$. In physics language, these sectors are labelled by quasi-momentum k .

Remark. On the infinite lattice, the states $|k\rangle$ are not normalizable vectors in $\ell^2(\mathbb{Z})$. They are generalized eigenvectors, just like plane waves on $\mathbb{R}$. One can make everything completely rigorous by first considering a finite periodic lattice with N sites and then taking $N \rightarrow \infty$.

Part (b): eigenvectors of the translation operator

The irreducible unitary representations of $\mathbb{Z}$ are one-dimensional. They are characters

$$\chi_k : \mathbb{Z} \longrightarrow U(1), \quad \chi_k(n) = e^{ikn},$$

where $k \in [-\pi, \pi)$. The parameter k is defined modulo 2π , which is why this interval is a fundamental domain.

The sheet defines

$$|k\rangle = \sum_{n \in \mathbb{Z}} e^{ikn} |n\rangle.$$

Using $T|n\rangle = |n+1\rangle$, we compute

$$T|k\rangle = \sum_{n \in \mathbb{Z}} e^{ikn} T|n\rangle = \sum_{n \in \mathbb{Z}} e^{ikn} |n+1\rangle.$$

Now re-index the sum by setting $m = n+1$, so $n = m-1$. Then

$$T|k\rangle = \sum_{m \in \mathbb{Z}} e^{ik(m-1)} |m\rangle = e^{-ik} \sum_{m \in \mathbb{Z}} e^{ikm} |m\rangle = e^{-ik} |k\rangle.$$

Sign convention warning. With the convention $T|n\rangle = |n+1\rangle$ and $|k\rangle = \sum_n e^{ikn} |n\rangle$, the eigenvalue is e^{-ik} , not e^{ik} . To obtain the eigenvalue e^{ik} , one may instead define

$$|k\rangle_- := \sum_{n \in \mathbb{Z}} e^{-ikn} |n\rangle.$$

Then

$$T|k\rangle_- = e^{ik} |k\rangle_-.$$

This is only a sign convention for k , and it does not change the energy $E(k) = -2t \cos k$, since $\cos(-k) = \cos k$.

Thus the Bloch states diagonalize the translation operator. Depending on the chosen sign convention, the character is either e^{ik} or e^{-ik} . The physics is unchanged by replacing k with $-k$.

Part (c): diagonalizing the Hamiltonian

We now compute $H|k\rangle$. Using

$$H|n\rangle = -t(|n-1\rangle + |n+1\rangle),$$

we get

$$\begin{aligned} H|k\rangle &= \sum_{n \in \mathbb{Z}} e^{ikn} H|n\rangle \\ &= -t \sum_{n \in \mathbb{Z}} e^{ikn} (|n-1\rangle + |n+1\rangle). \end{aligned}$$

We treat the two sums separately.

For the first sum, set $m = n-1$, so $n = m+1$. Then

$$\sum_{n \in \mathbb{Z}} e^{ikn} |n-1\rangle = \sum_{m \in \mathbb{Z}} e^{ik(m+1)} |m\rangle = e^{ik} \sum_{m \in \mathbb{Z}} e^{ikm} |m\rangle = e^{ik} |k\rangle.$$

For the second sum, set $m = n + 1$, so $n = m - 1$. Then

$$\sum_{n \in \mathbb{Z}} e^{ikn} |n + 1\rangle = \sum_{m \in \mathbb{Z}} e^{ik(m-1)} |m\rangle = e^{-ik} \sum_{m \in \mathbb{Z}} e^{ikm} |m\rangle = e^{-ik} |k\rangle.$$

Therefore

$$\begin{aligned} H|k\rangle &= -t(e^{ik} + e^{-ik})|k\rangle \\ &= -2t \cos k |k\rangle. \end{aligned}$$

Thus $|k\rangle$ is an eigenvector of H with eigenvalue

$$\boxed{E(k) = -2t \cos k.}$$

This function $E(k)$ is the dispersion relation. It tells us the energy as a function of quasi-momentum.

Several features are worth noting:

- The energy is periodic:

$$E(k + 2\pi) = E(k).$$

This reflects the fact that k is defined modulo the reciprocal lattice.

- The minimum energy is

$$E(0) = -2t.$$

- The maximum energy is

$$E(\pi) = 2t.$$

- The bandwidth is

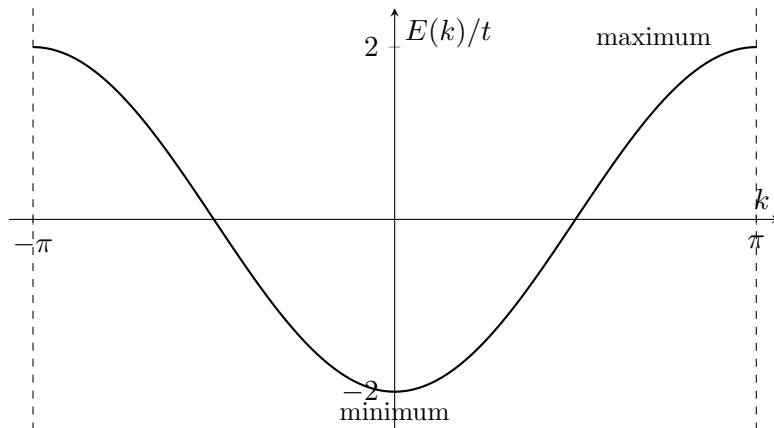
$$2t - (-2t) = 4t.$$

Part (d): plot of the band

In units where energy is measured in multiples of t , the dispersion is

$$\frac{E(k)}{t} = -2 \cos k.$$

For $k \in [-\pi, \pi]$, the plot is:



The tight-binding band $E(k) = -2t \cos k$ on the Brillouin zone $[-\pi, \pi]$.

The curve is a single energy band. Translation symmetry allowed us to reduce an infinite-dimensional eigenvalue problem to a one-parameter family of one-dimensional problems labelled by k .